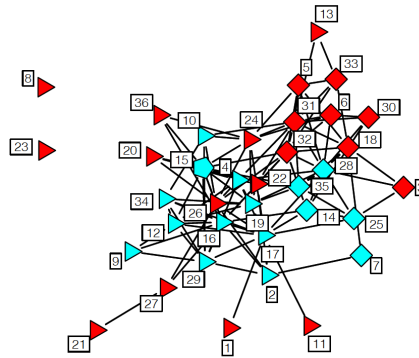


L13: Link Prediction

Let us introduce the Link Prediction problem with an example. The following network refers to a set of 36 lawyers (partners and associates) working for a law firm in New England. Two lawyers are connected with an edge if they indicated (through a survey) that they have worked together in a case. We know several attributes for each lawyer: seniority in the firm (indicated by the number next to each node), gender (nodes 27, 29, 34 are females), office location (indicated by the shape of the node -- there are three locations in the dataset), and type of practice (red for Litigation and cyan for Corporate Law).



Source: "Statistical analysis of network data" by E.D.Kolaczyk. <http://math.bu.edu/ness12/ness2012-shortcourse-kolaczyk.pdf>

Suppose that we can observe a portion of this graph -- but not the whole thing. *How would you infer whether two nodes that appear disconnected are actually connected or not?* Intuitively, you can rely on two sources of data:

- The first is the node attributes -- for instance, it may be more likely for two lawyers to work together if they share the same office location and they both practice corporate law.
- The second is the topological information from the known edges. For instance, if we know that A and B are two nodes that do not share any common neighbors, it may be unlikely that A and B are connected.

Let us start by stating an important assumption. In the following, we assume that the missing edges are *randomly missing* -- so, whether an edge is observed or not does *not* depend on its own attributes. Without this assumption, the problem is significantly harder (imagine solving the link prediction problem in the context of a social network in which certain kinds of relationships are often hidden).

Let us first define some topological metrics that we can use as "predictor variables" or "features" in our statistical model. Consider a node i , and let N_i^{obs} be the set of its observed neighbors. As we have seen, many networks in practice are highly clustered. This means that if two nodes have highly overlapping observed neighbors, they are probably connected as well.

A commonly used metric to quantify this overlap for nodes i and j is the size of the intersection $|N_i^{obs} \cap N_j^{obs}|$. The normalized version of this metric is the *Jaccard similarity*,

$$s(i, j) = \frac{|N_i^{obs} \cap N_j^{obs}|}{|N_i^{obs} \cup N_j^{obs}|}$$

which is equal to one if the two nodes i and j have identical observed neighbors.

Another topological similarity metric, more relevant to network analysis, is the following:

$$s(i, j) = \sum_{k \in N_i^{obs} \cap N_j^{obs}} \frac{1}{\log |N_k^{obs}|}$$

Here, node k is a common neighbor of both i and j . The idea is that if k is highly connected to other nodes, it does not add much evidence for a connection between i and j . If k is only connected to i and j though, it makes that connection more likely. This metric is sometimes referred to as "Adamic-Adar similarity".

There are several more topological similarity metrics in the literature -- but the previous two give you the basic idea.

Together with topological similarity scores $s(i,j)$, we can also use the node attributes to construct additional predictor variables for every node pair. Returning to the lawyer collaboration example, we could define, for instance, the following five variables:

$$Z_{i,j}^{(1)} = \text{seniority}_i + \text{seniority}_j$$

$$Z_{i,j}^{(2)} = \text{practice}_i + \text{practice}_j$$

$$Z_{i,j}^{(3)} = 1 \text{ if } (\text{practice}_i = \text{practice}_j) \text{ and } 0 \text{ otherwise}$$

$$Z_{i,j}^{(4)} = 1 \text{ if } (\text{gender}_i = \text{gender}_j) \text{ and } 0 \text{ otherwise}$$

$$Z_{i,j}^{(5)} = 1 \text{ if } (\text{office}_i = \text{office}_j) \text{ and } 0 \text{ otherwise}$$

We can also represent with the same notation any topological similarity score between node pairs. For instance,

$$Z_{i,j}^{(6)} = |N_i^{obs} \cap N_j^{obs}|$$

Many other such predictor variables can be defined, based on the node attributes and topological similarity metrics.

Now that we have defined these predictor variables for each pair of nodes, we can design a *binary classifier* using *Logistic Regression*.

Suppose that $\mathbf{Y}$ is the (complete) adjacency matrix of the network. The binary variables we want to model statistically are the adjacency matrix elements $\mathbf{Y}_{i,j}$ (equal to one if there is a connection and zero otherwise). Our training dataset consists of the set of *observed* node pairs $\mathbf{Y}^{obs}$ -- these are the node pairs for which we know whether there is an edge ($\mathbf{Y}_{i,j}^{obs} = 1$) or not ($\mathbf{Y}_{i,j}^{obs} = 0$). The remaining node pairs of $\mathbf{Y}$ are represented by $\mathbf{Y}^{miss}$ and we do not know whether they are connected ($\mathbf{Y}_{i,j}^{miss} = 1$) or not ($\mathbf{Y}_{i,j}^{miss} = 0$).

The equation that defines the Logistic Regression model is:

$$\log \left[\frac{P(Y_{i,j}=1|\mathbf{Z}_{i,j}=\mathbf{z})}{P(Y_{i,j}=0|\mathbf{Z}_{i,j}=\mathbf{z})} \right] = \beta^T \mathbf{z}$$

where $\mathbf{Y}_{i,j} = 1$ means that the edge between nodes i and j actually exists (observed or missing) -- while $\mathbf{Y}_{i,j} = 0$ means that the edge does *not* exist. The vector $\mathbf{z}$ includes all the predictor variables for that node pair (we defined six such variables above). Finally, the vector β is the vector of the regression coefficients, and *it is assumed to be the same for all node pairs*.

A logistic regression model can be trained based on the observed data $\mathbf{Y}^{obs}$, calculating the optimal vector of regression coefficients β . If you want to learn how this optimization is done computationally, please refer to any machine learning or statistical inference textbook.

After training the model with the observed node pairs (connected or not), we can use the logistic regression model to predict whether any missing node pair is actually connected using the following equation:

$$P(Y_{i,j}^{miss} = 1 | \mathbf{Z}_{i,j} = \mathbf{z}) = \frac{e^{\beta^T \mathbf{z}}}{1 + e^{\beta^T \mathbf{z}}}$$

This equation follows directly from the logistic regression model. For instance, if the previous probability is larger than 0.5 for a node pair (i,j) , we can infer that the two nodes are connected.

The link prediction problem is very general and any other binary classification algorithm could be used instead of logistic regression. For example, we could use a support vector machine (SVM) or a neural network.

Food For Thought

In the link prediction framework we presented here, the logistic regression coefficients are the same for every node pair. What does this mean/assume about the structure of the network? How would you compare this approach with link prediction using the HRG modeling approach we studied in Lesson-12?